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# A hybrid architecture with separable convolutions and attention for lung and colon cancer detection

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ABSTRACT

Early and accurate detection of lung and colon cancer is critical for improving patient survival rates. However, conventional classification techniques often struggle with the complexity of medical imaging data. In this study, we propose a novel hybrid architecture, the Separable Convolution and Attention Mechanism (SCA-mechanism), to enhance the accuracy of cancer classification. The proposed model integrates separable convolutions with residual connections to facilitate efficient feature extraction while incorporating attention mechanisms to emphasize key regions indicative of malignancy. We evaluated the performance of the SCA-mechanism model against state-of-the-art deep learning techniques using a benchmark dataset of lung and colon cancer images. Experimental results demonstrate that our proposed approach achieves superior classification performance, outperforming conventional deep learning methods in both accuracy and computational efficiency. These findings highlight the potential of the SCA-mechanism as a promising tool for automated cancer diagnosis, offering significant advancements in computationally efficient medical applications and clinical decision making.

## 1. Introduction

Lung and colon cancers are two of the most prevalent and life-threatening malignancies globally, accounting for more than two million deaths annually [1]. Early and accurate diagnosis plays a crucial role in improving patient prognosis, as the five-year survival rate can exceed 90% when detected at an early stage, in stark contrast to less than 20% for late-stage diagnoses [2]. This critical disparity has catalyzed the development of automated and scalable diagnostic solutions that can assist clinicians in high-throughput, high-stakes decision-making environments.

Histopathological image analysis remains the gold standard for confirming cancer diagnoses. These high-resolution microscopy images contain intricate tissue structures and cellular details that are indispensable for detecting malignancies. However, the analysis process is often hindered by domain-specific challenges such as staining variability, high inter-class similarity, intra-class heterogeneity, and subtle morphological variations. Manual interpretation by expert pathologists is time-consuming, subject to fatigue and variability, and difficult to scale. While traditional machine learning methods have attempted to assist in diagnostic interpretation, their dependency on handcrafted features and shallow representation learning often limits their ability to generalize across datasets and cancer types [3].

Deep learning–especially convolutional neural networks (CNNs)–has revolutionized medical image analysis by enabling end-to-end feature extraction and classification from raw pixel data [4]. Nevertheless, most conventional CNN architectures are computationally intensive and require substantial resources for training and inference, especially when dealing with ultra-high-resolution histopathological images. Moreover, CNNs pretrained on natural image datasets may fail to capture domain-specific patterns that are critical in differentiating benign from malignant tissues. These limitations constrain the deployment of deep learning models in computationally efficient or resource-constrained clinical settings.

To address these challenges, we propose the SCA-mechanism—a hybrid and computationally efficient deep learning architecture tailored for histopathological cancer classification. The model combines three key innovations: (1) *Separable Convolutions*, which decouple spatial and channel-wise processing to significantly reduce parameter count and training cost; (2) *Residual Connections*, which mitigate vanishing gradient issues and enable stable training of deeper networks; and (3)

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*Attention Mechanisms*, which dynamically prioritize clinically relevant features such as abnormal nuclei or glandular structures. Together, these components allow the SCA-mechanism to capture fine-grained visual cues while maintaining a lightweight and interpretable design.

We validate the model on the publicly available LC25000 dataset, which contains 25,000 histopathological images evenly distributed across five classes, including both benign and malignant lung and colon tissue categories. The proposed architecture achieves a classification accuracy of 98.82%, surpassing several state-of-the-art CNN models in both predictive performance and computational efficiency. Notably, our model reduces average training time per epoch by nearly half compared to widely adopted architectures such as VGG16, ResNet50, and EfficientNetB7. This performance advantage positions the SCA-mechanism as a practical candidate for integration into edge computing devices or computationally efficient clinical diagnostic pipelines.

**The major contributions of this study are summarized as follows:**

1. We introduce a novel hybrid architecture (SCA-mechanism) that integrates separable convolutions, residual learning, and attention mechanisms to enhance classification accuracy and interpretability in histopathological cancer detection.
2. The model achieves 98.82% classification accuracy on the LC25000 dataset while reducing training time significantly compared to conventional deep learning architectures.
3. The design supports computationally efficient deployment and efficient inference, making it suitable for computationally efficient diagnostic systems and resource-limited clinical environments.

The rest of this paper is structured as follows: Section 2 presents related works and identifies key research gaps. Section 3 describes the dataset and preprocessing pipeline. Section 4 elaborates on the proposed model architecture. Section 5 details the experimental setup, results, and comparative evaluations. The key findings and current limitations are discussed in Section 6. Finally, Section 7 concludes the study and outlines avenues for future research.

## 2. Literature review

Recent advancements in deep learning have significantly influenced the field of cancer detection, particularly in the classification of histopathological images for colon and lung cancer. However, a number of technical and clinical challenges persist, including limited generalization, computational inefficiency, and lack of interpretability, all of which impact the feasibility of computationally efficient deployment in healthcare settings. This section reviews recent works in two major application areas–colon cancer and lung cancer detection–and outlines their limitations in contrast to the design motivations of the proposed SCA-mechanism.

### 2.1. Deep learning for colon cancer detection

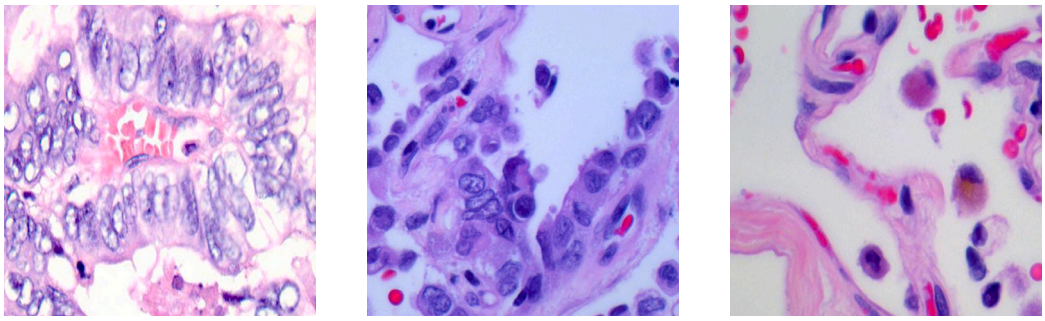
In colon cancer detection, Khan et al. [5] introduced a Swin Transformer-based model using the LC25000 dataset, which achieved high binary classification accuracy. However, the model's performance dropped notably under multiclass conditions due to its reliance on local attention windows, which limited its ability to capture global contextual features. Alsubai et al. [6] combined Inception-ResNetV2 with Local Binary Pattern (LBP) features to leverage texture descriptors, but the handcrafted LBP component rendered the model sensitive to noise and resolution inconsistencies often present in real-world histopathological images. Sobur et al. [7] employed image augmentation techniques on a CNN model to boost diversity [8,9] yet such transformations risk distorting structural tissue features and may inadvertently hinder interpretability or generalization. Mohamed et al. [10] applied a Fisher Mantis Optimizer to fine-tune CNN weights, achieving competitive accuracy but at the expense of training complexity and scalability, particularly for time-constrained clinical environments [11]. Karthikeyan et al. [12] proposed a CNN pipeline with a ranking module trained on a very small dataset (284 training samples), which severely limited the model's generalizability to larger or more diverse populations. Across these approaches, common limitations include overreliance on handcrafted components, poor scalability to multiclass classification tasks, and sensitivity to dataset quality—particularly in terms of class balance and image variation. These shortcomings motivate a need for end-to-end learnable, computationally efficient architectures that remain robust across variable imaging conditions and can be deployed in resource-constrained environments.

### 2.2. Deep learning for lung cancer detection

In the domain of lung cancer detection, Yao et al. [13] implemented a nnU-Net framework incorporating a segmentation-classification pipeline based on contrast-enhanced CT images. Although effective, the model required complex pre- and post-processing, making it less suitable for rapid or scalable deployment. Sangeetha et al. [14] proposed a multimodal fusion deep neural network (MFDNN) that achieved high accuracy by combining multiple imaging modalities, yet its dependency on synchronized multimodal data complicates generalization and reduces feasibility for hospitals lacking such infrastructure. Wani et al. [15] developed DeepXplainer–a two-stage CNN-XGBoost hybrid that enhanced interpretability–but this came at the cost of increased inference time, which is critical in clinical workflows requiring computationally efficient decision support [16]. Gayap and Akhloufi [17] surveyed major CNN and ViT-based frameworks, identifying significant issues in model reproducibility due to dataset overfitting and lack of standardized evaluation benchmarks. Deshpande et al. [18] proposed a hybrid system combining handcrafted radiomic features and deep learning for lung nodule classification, improving performance on specific datasets like LIDC-IDRI. However, their approach necessitated domain-specific feature engineering, limiting its adaptability across institutions with varying imaging standards. These studies highlight persistent issues such as architectural complexity, interpretability-efficiency trade-offs, dependence on handcrafted or modality-specific features, and poor real-world adaptability—especially when imaging artifacts, resolution inconsistencies, or limited resources are involved.

### 2.3. Computationally efficient deployment and clinical gaps

Despite promising results in research environments, many of the models reviewed face critical challenges in real-world deployment. Deep networks such as Swin Transformer [19] often incur significant latency, making them impractical for computationally efficient clinical use, especially in screening workflows or resource-limited hospitals. Moreover, image quality variations–stemming from different scanning devices, staining protocols, and sample preparation–pose significant risks to models trained on highly curated datasets. Handcrafted features (e.g., LBP or radiomics) are often dataset-specific and non-transferable across institutions [20]. Most notably, interpretability is often treated as an afterthought; either absent altogether or appended post hoc, limiting transparency and clinician trust [21]. The lack of integrated attention or explanation modules reduces the utility of AI systems in high-stakes decisions where visual rationale is crucial. Finally, dataset-related challenges such as class imbalance, small sample sizes, or homogeneous imaging sources exacerbate performance variance and weaken model generalization—issues that are rarely addressed systematically across studies.



(a) Colon adenocarcinoma (colon_aca) (b) Lung adenocarcinoma (lung_aca) (c) Benign lung tissue (lung_n)

**Fig. 1.** Sample images from the Lung and Colon Cancer Histopathological Image Dataset (LC25000).

### 2.4. Research gaps and mitigation

Across the literature, there remains a clear need for a unified, lightweight architecture that overcomes the combined challenges of interpretability, efficiency, and robustness under real-world imaging variability. Existing approaches either focus on accuracy at the expense of scalability (e.g., complex transformer models), or on handcrafted feature enrichment without addressing deployment bottlenecks. In contrast, the proposed **SCA-mechanism** responds directly to these research gaps by integrating Separable Convolutions for computational efficiency, Residual Connections for gradient stability and generalization, and Attention Mechanisms for built-in interpretability.

## 3. Data description and pre-processing

### 3.1. Data description

For this study, we utilized the Lung and Colon Cancer Histopathological Image Dataset (LC25000), a well-structured and widely used benchmark dataset for cancer classification tasks. The dataset consists of 25,000 high-resolution histopathology images, each with a resolution of 768 × 768 pixels in JPEG format. It is systematically categorized into five distinct classes, with 5000 images per class, representing both malignant and benign tissue samples. The five classes include:

- Lung adenocarcinoma (lung_aca)—Malignant lung tissue.
- Lung squamous cell carcinoma (lung_scc)—A subtype of malignant lung tissue.
- Benign lung tissue (lung_n)—Normal lung tissue.
- Colon adenocarcinoma (colon_aca)—Malignant colon tissue.
- Benign colonic tissue (colon_n)—Normal colon tissue.

This dataset serves as a valuable resource for training and evaluating machine learning models for cancer detection, given its high-quality annotations and balanced distribution of classes.

### 3.2. Data preprocessing

We made a few pragmatic choices in preprocessing to keep the pipeline fast, reproducible, and fair to all baselines, while still respecting the fine detail in histopathology images. Below we explain each step and the trade-offs we considered.

- **Image resizing:** All images are resized to 224 × 224 to match the network input. We are aware that downscaling can soften very small textures (e.g., nuclear chromatin). To mitigate this, we use anti-aliased bilinear interpolation and avoid aggressive cropping so that nuclei and gland boundaries remain visible. This size strikes a practical balance between detail and throughput; in future work, a simple extension is to add a multi-scale pathway (e.g., larger inputs or image pyramids) to further protect fine-grained cues.
- **Normalization:** Our default is per-pixel min–max scaling to $[0, 1]$ (divide by 255), which is simple and stable across models. An equally reasonable alternative is per-channel standardization (mean subtraction and standard-deviation scaling) computed *only on the training split* to avoid leakage. We keep min–max as the default for comparability, and note that switching to z-score normalization does not change the rest of the pipeline.
- **Data splitting:** We explicitly use three stratified splits to aid model tuning and guard against overfitting:
  - **Training (80%)**—used to learn parameters,
  - **Validation (10%)**—used for hyperparameter selection, early stopping, and learning-rate scheduling,
  - **Test (10%)**—held out once for the final report.

  The validation set provides a clean signal for stopping before overfitting and choosing sensible training settings.
- **Data augmentation:** We apply light, morphology-preserving transforms on the fly:
  - Random rotations within ±15°—a conservative range that captures typical slide orientation changes without bending tissue structures,
  - Horizontal and vertical flips—to reflect lateral and axial variability.
  - Bounded zoom ($[0.9, 1.1]$) or small random resized crops (area $[0.9, 1.0]$) was enabled to add scale invariance when needed, while keeping distortions controlled.

  These choices increase diversity without introducing artifacts that could mislead the model.
- **Shuffling and class balance:** To avoid order effects and maintain stable gradients, we shuffle samples every epoch *within each class* and construct mini-batches that respect class proportions (i.e., batches remain approximately balanced). This prevents any class from dominating updates, especially early in training. All random operations are seeded for reproducibility.

Thus, the pipeline preserves as much detail as practical at 224 × 224, offers a clear path to alternative normalizations, uses a dedicated validation split for tuning, keeps augmentations deliberately mild to match pathology practice, and maintains balanced, shuffled batches for reliable optimization.

## 4. Proposed methodology

This section presents the design and implementation of the Hybrid Architecture Combining Separable Convolutions and Attention Mechanisms (SCA-mechanism), developed for the automated diagnosis of

lung and colon cancer. Our approach integrates advanced convolutional techniques with attention mechanisms to enhance feature extraction and emphasize clinically significant regions in medical images. The architectural design prioritizes both high classification accuracy and computational efficiency, ensuring its suitability for computationally efficient diagnostic applications.

### 4.1. SCA-mechanism architecture

To effectively process and analyze medical images for lung and colon cancer detection, we developed the SCA-mechanism model. The architecture is composed of four key components:

- **Input Layer:** Accepts a preprocessed medical image with dimensions $224 \times 224 \times 3$.
- **Feature Extraction Blocks:** Utilize separable convolutional layers with increasing filter depths to efficiently extract hierarchical features.
- **Attention Mechanism:** Enhances the model's ability to focus on critical regions in the image, improving feature discrimination.
- **Classification Head:** Outputs the final prediction by mapping extracted features to cancer classification labels.

Fig. 2 provides a schematic representation of the SCA-mechanism architecture, illustrating the flow of data between these components.

### 4.2. Feature extraction pipeline

The SCA-mechanism processes input medical images through a structured pipeline to achieve precise and reliable cancer classification. Initially, a preprocessed histopathology image (Fig. 1(a)) with dimensions $224 \times 224 \times 3$ is fed into the model. The feature extraction process consists of three successive feature extraction blocks, each designed to capture multi-scale hierarchical information. The architecture employs:

- Separable Convolutional Layers to reduce computational complexity while preserving spatial and depth-wise feature representations.
- Progressively increasing filters to enhance feature granularity:
  - Block 1: 32 filters to capture low-level features.
  - Block 2: 64 filters for mid-level feature extraction.
  - Block 3: 128 filters to extract high-level features.
- Residual Connections to facilitate gradient flow and improve training stability.

By integrating these architectural components, the SCA-mechanism model achieves an optimal balance between computational efficiency and classification accuracy, making it a scalable and clinically viable solution for lung and colon cancer detection.

### 4.3. Separable convolutions with residual connections

The proposed SCA-mechanism combines depthwise–separable convolutions (for low-cost spatial filtering), pointwise projections (for channel mixing), residual connections (for stable optimization), and lightweight channel attention (for feature recalibration). Compared with conventional backbones (e.g., VGG/ResNet/EfficientNet), SCA targets a favorable accuracy–efficiency trade-off suitable for histopathology tiles: separable convolutions reduce parameters and FLOPs, residual links preserve gradient flow and mitigate degradation, and attention emphasizes diagnostically salient responses (e.g., nuclei and gland boundaries) without incurring heavy computational overhead. In ablation study, we quantify the contribution of each component and compare against alternative design choices (dilated convolution, pooling variants, optimizers, and regularization levels).

#### 4.3.1. Separable convolutions

Separable convolutions decompose standard convolution operations into depthwise and pointwise convolutions, significantly reducing the number of parameters and computational overhead while preserving feature extraction capabilities. This makes them particularly advantageous for processing high-resolution histopathological images in computationally efficient diagnostic applications.

A conventional convolution operation is mathematically expressed as follows [22]:

$$\boldsymbol{Y} = \boldsymbol{I} \cdot \boldsymbol{K} + \boldsymbol{b} \tag{1}$$

where $\boldsymbol{I}$ represents the input image, $\boldsymbol{K}$ denotes the convolutional kernel, $\boldsymbol{b}$ is the bias term, and $(\cdot)$ signifies the convolution operation.

In contrast, a depthwise separable convolution divides this operation into two sequential steps:

1. **Depthwise Convolution:** This step applies individual convolutional filters to each input channel separately, capturing spatial features independently. Given an input with $C$ channels, depthwise convolution [23] applies $C$ distinct kernels $\boldsymbol{K}_c$, producing intermediate feature maps $\boldsymbol{D}_c$:

$$\boldsymbol{D}_c = \boldsymbol{I}_c \cdot \boldsymbol{K}_c, \quad \forall c \in \{1, 2, \ldots, C\} \tag{2}$$

2. **Pointwise Convolution:** This step employs $1 \times 1$ convolutions to integrate the depthwise convolution outputs across channels, yielding the final feature maps $\boldsymbol{Y}$ [24]:

$$\boldsymbol{Y} = \sum_{c=1}^{C} \boldsymbol{D}_c \cdot \boldsymbol{W}_c \tag{3}$$

where $\boldsymbol{W}_c$ represents the pointwise convolutional weights.

#### 4.3.2. Alternative convolutional and pooling choices

To probe receptive-field design, we evaluate a variant that replaces the middle separable block with a dilated (rate = 2) separable convolution (**SCA Dilated**). This modification enlarges the effective field-of-view without additional downsampling, which can better capture glandular context. We also compare global average pooling (GAP) with global max pooling (GMP). GAP tends to yield smoother, more stable gradients and improves calibration; GMP can highlight sparse, high-activation foci but was less robust to noise. The results are reported in the ablation study section.

#### 4.3.3. Advantages of separable convolutions in medical imaging

Separable convolutions are particularly well-suited for histopathological image analysis, offering the following benefits:

- Reduced Computational Complexity: By replacing standard convolutions with depthwise and pointwise operations, separable convolutions significantly lower the number of trainable parameters, reducing memory requirements and increasing processing efficiency.
- Efficient Feature Extraction: These convolutions retain fine-grained spatial information essential for detecting subtle cancerous abnormalities, making them ideal for histopathological classification.
- Enhanced Scalability: The lower computational overhead makes separable convolutions suitable for computationally efficient diagnostic applications, enabling deployment in resource-constrained clinical settings.

By integrating separable convolutions with residual connections, our architecture optimally balances accuracy and efficiency, ensuring robust cancer detection while maintaining computational feasibility.

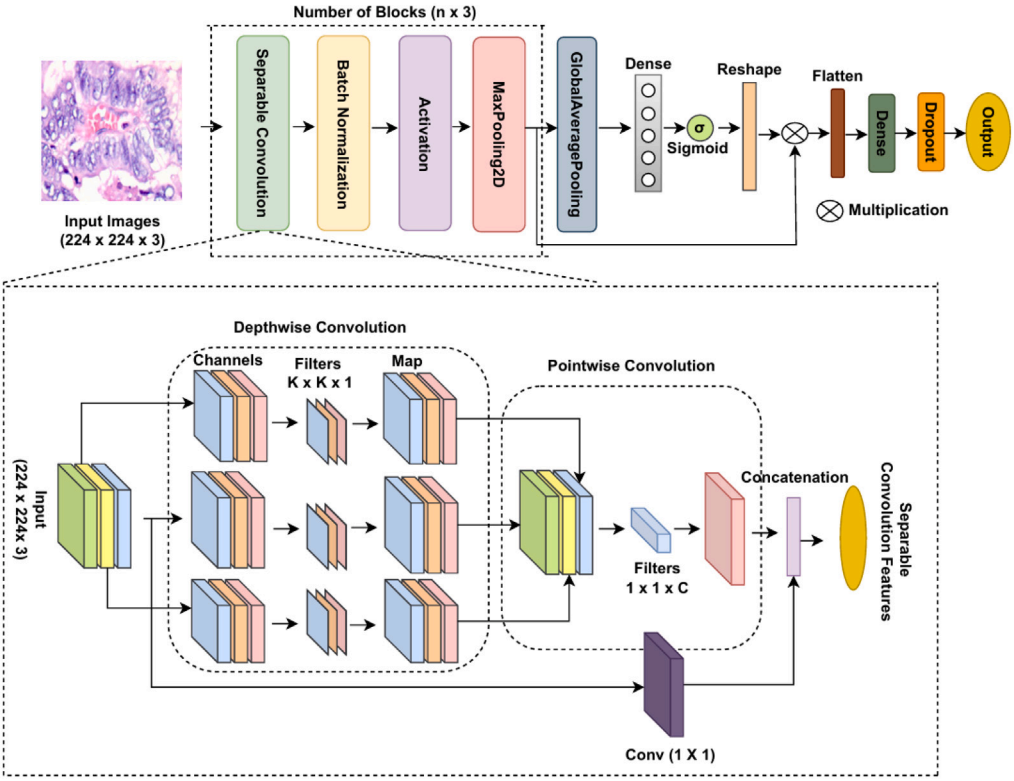



**Fig. 2.** The proposed SCA-mechanism architecture framework, illustrating the sequential flow from input processing through feature extraction, integration of attention mechanisms, and final classification head for multi-class classifications.

#### 4.3.4. Residual connections and attention

Residual connections play a crucial role in deep learning architectures by mitigating the vanishing gradient problem, facilitating the training of deeper networks. In our model, each separable convolutional block incorporates a residual path, allowing the input to bypass the main transformation layers and be directly added to the output. When necessary, a $1 \times 1$ convolution is used to align dimensions before the addition operation, ensuring compatibility.

We therefore contrast *SepConv+Attn (no residual)* with *SepConv+Attn (Residual)* to isolate the stability and accuracy gains. Channel attention is implemented as a lightweight squeeze–excite gating, promoting discriminative channels (e.g., nuclear chromatin patterns) while suppressing distractors. The net effect on macro-F1/AUC and calibration is summarized in the ablation study section.

Mathematically, the output $\boldsymbol{Y}$ of a residual block is defined as [25]:

$$\boldsymbol{Y} = \boldsymbol{F}(\boldsymbol{X}, \boldsymbol{W}_i) + \boldsymbol{X} \tag{4}$$

where:

- $\boldsymbol{X}$ represents the input to the block.
- $\boldsymbol{F}(\boldsymbol{X}, \boldsymbol{W}_i)$ denotes the transformation function applied by the separable convolutional layers, including convolutions, activation functions, and normalization.
- If $\boldsymbol{F}(\boldsymbol{X}, \boldsymbol{W}_i)$ and $\boldsymbol{X}$ have mismatched dimensions, a $1 \times 1$ convolution is applied to $\boldsymbol{X}$ to ensure dimensional consistency before addition.

#### 4.3.5. Advantages of residual connections in deep networks

The inclusion of residual connections provides several key advantages:

- Improved Gradient Flow: By allowing gradients to propagate more effectively through the network, residual connections prevent degradation in deeper architectures and enhance learning efficiency.
- Enhanced Model Stability: They facilitate faster and more stable convergence by preserving important low-level features while enabling deeper feature extraction.
- Better Generalization: The ability to train deeper models without excessive overfitting improves the model's capacity to learn complex patterns in medical images.
- Scalability for Deeper Architectures: Residual learning allows the network to scale efficiently, enabling the construction of highly expressive models while maintaining computational feasibility.

By incorporating residual connections into our SCA-mechanism, we enhance both training efficiency and model performance, enabling deeper architectures that maintain high accuracy while overcoming traditional training challenges.

### 4.4. Attention mechanism

To enhance the model's ability to focus on clinically relevant regions within medical images, we integrated an Attention Mechanism following the feature extraction layers. This module helps the network prioritize critical features, thereby improving classification sensitivity and accuracy. The attention mechanism operates through the following sequential stages:

#### 4.4.1. Global Average Pooling (GAP)

Global Average Pooling (GAP) is employed to aggregate spatial information across the feature maps, producing a channel-wise descriptor $\boldsymbol{Z}$ that encodes global contextual information [26]. This step effectively reduces the spatial dimensions while retaining essential information:

$$Z_c = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} X_{c,i,j} \tag{5}$$

where:

- $\boldsymbol{H}$ and $\boldsymbol{W}$ denote the height and width of the feature maps.
- $\boldsymbol{X}_{c,i,j}$ represents the activation at channel $c$ and spatial position $(i, j)$.

This operation reduces feature dimensionality, enabling the model to learn globally significant patterns while preserving critical class-specific details.

#### 4.4.2. Fully connected layers for attention weights

The channel-wise descriptor $\boldsymbol{Z}$ is then processed through two fully connected layers, where attention weights $\boldsymbol{s}$ are computed to determine the relative importance of each feature map [27]:

$$\boldsymbol{s} = \sigma(\boldsymbol{W}_2 \cdot \delta(\boldsymbol{W}_1 \cdot \boldsymbol{Z} + \boldsymbol{b}_1) + \boldsymbol{b}_2) \tag{6}$$

where:

- $\delta$ represents the ReLU activation function.
- $\sigma$ is the sigmoid activation function.
- $\boldsymbol{W}_1, \boldsymbol{W}_2$ are the learnable weights of the fully connected layers.
- $\boldsymbol{b}_1, \boldsymbol{b}_2$ are the bias terms.

This step non-linearly transforms the feature descriptors, allowing the model to learn complex feature interactions and assign higher importance to key diagnostic patterns.

#### 4.4.3. Feature recalibration

The computed attention weights $\boldsymbol{s}$ are then reshaped and applied element-wise to the original feature maps $\boldsymbol{X}$ to obtain the recalibrated feature maps $\boldsymbol{X}'$, effectively enhancing the most relevant features while suppressing less informative ones [28]:

$$\boldsymbol{X}'_{c,i,j} = \boldsymbol{X}_{c,i,j} \times \boldsymbol{s}_c \tag{7}$$

where:

- $\boldsymbol{X}_{c,i,j}$ represents the original feature maps.
- $\boldsymbol{s}_c$ is the computed attention weight for channel $c$.

#### 4.4.4. Impact on cancer detection

By emphasizing diagnostically significant regions in histopathological images, this attention mechanism improves classification performance in several ways:

- Enhanced Sensitivity to Malignant Features: The model learns to focus on cancerous regions, improving its ability to distinguish between benign and malignant tissues.
- Improved Interpretability: Attention-based feature recalibration provides explainability, making it easier to identify which regions contribute to the final classification decision.
- Better Generalization: By dynamically adjusting feature importance, the network adapts to variations in histopathological images, reducing overfitting.

Integrating this attention mechanism into our SCA-mechanism architecture significantly boosts diagnostic accuracy, making the model more reliable for real-world medical applications.

### 4.5. Classification head

Following feature extraction and attention-based refinement, the Classification Head is responsible for categorizing input images into predefined classes, such as lung cancer and colon cancer subtypes. The classification process consists of the following key steps:

#### 4.5.1. Flattening

The recalibrated feature maps $\boldsymbol{X}'$ are transformed into a one-dimensional feature vector $\boldsymbol{v}$, ensuring compatibility with fully connected layers [29]:

$$\mathbf{v} = \text{Flatten}(\mathbf{X}') \tag{8}$$

#### 4.5.2. Fully connected layer

The flattened vector $\boldsymbol{v}$ is passed through a dense layer with 256 units, ReLU activation, and L2 regularization to learn complex feature representations [30]:

$$\boldsymbol{h} = \delta(\boldsymbol{W}_3 \cdot \boldsymbol{v} + \boldsymbol{b}_3) + \lambda \|\boldsymbol{W}_3\|_2^2 \tag{9}$$

where:

- $\delta$ represents the ReLU activation function.
- $\boldsymbol{W}_3$ and $\boldsymbol{b}_3$ are the weights and biases of the dense layer.
- $\lambda$ is the L2 regularization parameter, which prevents overfitting.

#### 4.5.3. Dropout for regularization

To enhance generalization and mitigate overfitting, a dropout layer with a rate of 60% is applied to $\boldsymbol{h}$, randomly deactivating neurons during training [31]:

$$\mathbf{h}' = \text{Dropout}(0.6, \mathbf{h}) \tag{10}$$

#### 4.5.4. Output layer

The dropout-regularized vector $\mathbf{h}'$ is then passed through a softmax-activated dense layer, producing probabilistic predictions $\mathbf{p}$ for each class [32]:

$$p_k = \frac{\exp(\mathbf{W}_4^{(k)} \cdot \mathbf{h}' + b_4^{(k)})}{\sum_{m=1}^{M} \exp(\mathbf{W}_4^{(m)} \cdot \mathbf{h}' + b_4^{(m)})} \tag{11}$$

where:

- $M$ represents the total number of classes.
- $\mathbf{W}_4^{(k)}$ and $b_4^{(k)}$ are the weights and biases for class $k$.
- $p_k$ denotes the predicted probability for class $k$.

### 4.6. Training strategy

To ensure optimal performance and generalization, we adopted a comprehensive training strategy, incorporating appropriate loss functions, optimization techniques, and regularization mechanisms.

#### 4.6.1. Loss function

For multiclass classification, we employed categorical cross-entropy loss, which effectively measures the divergence between the true class labels and predicted probabilities [33]:

$$\iota = -\sum_{k=1}^{M} y_k \log(p_k) \tag{12}$$

where:

- $y_k$ is the ground truth label (1 for the correct class, 0 otherwise).
- $p_k$ is the predicted probability for class $k$.

#### 4.6.2. Optimizer

We employed the Adam optimizer due to its ability to adaptively adjust learning rates, ensuring efficient convergence [34]. To test optimizer sensitivity, we compare Adam, AdamW (decoupled weight decay), and RMSprop under identical schedules. We also study dropout rates $p \in \{0.3, 0.4, 0.6\}$ at the penultimate layer. The best setting is selected on the validation split and reported with mean ± SD over three seeds. Adam optimizer updates model parameters $\theta$ using the following mathematical backgrounds:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t \tag{13}$$

$$\upsilon_t = \beta_2 \upsilon_{t-1} + (1 - \beta_2) g_t^2 \tag{14}$$

$$\theta_t = \theta_{t-1} - \eta \frac{m_t}{\sqrt{\upsilon_t} + \epsilon} \tag{15}$$

where:

- $g_t$ represents the gradient at time step $t$.
- $\beta_1$ and $\beta_2$ are the exponential decay rates for first and second moment estimates.
- $\eta$ is the learning rate.
- $\epsilon$ is a small constant to prevent division by zero.

#### 4.6.3. Regularization techniques

To prevent overfitting and improve model generalization, we implemented the following regularization strategies:

- L2 Regularization: Applied within dense layers to penalize large weight magnitudes, discouraging overly complex models that may memorize the training data.
- Dropout: By randomly deactivating neurons during training, the dropout layer (60%) in the classification head encourages the model to learn robust, redundant representations, enhancing generalization.

#### 4.6.4. Robustness to noise and class imbalance

To mitigate label noise and class imbalance, we evaluate focal loss ($\gamma = 2$) and a class-balanced sampler. We additionally test label smoothing ($\varepsilon = 0.05$). These adjustments aim to stabilize training under minority classes and reduce overconfidence, particularly in borderline cases. This hybrid architecture offers a computationally efficient, scalable, and interpretable solution for lung and colon cancer diagnosis. Its ability to achieve high precision while maintaining computational economy makes it a strong candidate for computationally efficient clinical applications, assisting in early-stage cancer detection and decision-making processes.

## 5. Experimental result analysis

To train our model, we used Python 3.9.10 in the Kaggle environment, taking advantage of its powerful deep learning and scientific computing tools. We optimized performance with TensorFlow, running the training process on Kaggle's GPU-accelerated infrastructure (NVIDIA T4 16 GB). This setup significantly boosted processing efficiency, reducing training time and allowing for seamless execution of complex tasks. By leveraging Kaggle's free GPU resources, we successfully handled large-scale histopathological datasets while keeping computational demands manageable. The combination of TensorFlow's optimized configurations and Kaggle's robust cloud-based platform ensured smooth processing of high-resolution medical images, delivering optimal performance across all experiments.

The layer-wise architecture of the proposed SCA-mechanism model is presented in Table 1. This architecture integrates depthwise separable convolutions with an attention mechanism to enhance feature extraction while maintaining computational efficiency. The model consists of multiple convolutional layers followed by batch normalization, activation functions, and pooling layers to progressively refine the feature maps. The attention module, implemented via a combination of global average pooling, dense layers, and element-wise multiplication, strengthens relevant feature representations. Finally, fully connected layers classify the extracted features into the target categories.

### 5.1. Analysis of confusion matrices and ROC curves

To further analyze classification performance, Fig. 3 presents the confusion matrices for the SCA-mechanism model alongside six benchmark architectures: InceptionV3, MobileNetV2, DenseNet121, EfficientNetB7, VGG16, and ResNet50. These matrices provide a visual representation of the distribution of True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN), offering insights into each model's classification reliability.

- InceptionV3 (Fig. 3(a)): Achieves 66.19% test accuracy, but suffers from a high false negative rate, limiting its ability to detect malignant cases.
- MobileNetV2 (Fig. 3(b)): Performs better with 89.33% test accuracy, but remains prone to false positives and false negatives, reducing classification confidence.
- DenseNet121 (Fig. 3(c)): Shows 91.77% test accuracy, demonstrating improved precision and recall over MobileNetV2.
- EfficientNetB7 (Fig. 3(d)): Offers a more balanced classification with 96.71% test accuracy, effectively reducing misclassification errors.
- VGG16 and ResNet50 (Figs. 3(e) and 3(f)): Achieve 98.88% test accuracy, performing exceptionally well but still showing minor misclassifications.
- SCA-mechanism Model (Fig. 3(g)): Attains the highest test accuracy of 98.82%, significantly minimizing both false positives and false negatives, making it highly reliable for cancer classification.

The per-class evaluation presented in Table 2 shows that the SCA-mechanism achieves consistently high performance across all lung and colon cancer categories, with accuracy values ranging from 96.60% to 100%. Notably, the model attains perfect accuracy for normal lung tissue and near-perfect recognition of malignant colon samples, reflecting its strong ability to capture both fine-grained and high-level histopathological patterns. The overall macro-balanced accuracy of 98.82% further demonstrates the robustness of the proposed architecture across diverse tissue types.

Fig. 4 illustrates the Receiver Operating Characteristic (ROC) curves for the SCA-mechanism model and its benchmark counterparts. These curves provide a comprehensive evaluation of each model's ability to distinguish between malignant and benign cases, based on the trade-off between true positive rate (sensitivity) and false positive rate (1-specificity).

- InceptionV3 (Fig. 4(a)): Shows a moderate true positive rate, but struggles with false positives in certain regions.
- MobileNetV2 (Fig. 4(b)): Delivers reasonable performance, but is outperformed by more advanced models.
- DenseNet121 (Fig. 4(c)): Exhibits a higher sensitivity and specificity, surpassing earlier models.
- EfficientNetB7 (Fig. 4(d)): One of the top-performing transfer learning models, achieving improved true positive rates across all classes.
- VGG16 and ResNet50 (Figs. 4(e) and 4(f)): Show similar high-performance characteristics, with strong classification ability and low false positives.
- SCA-mechanism Model (Fig. 4(g)): Demonstrates an almost perfect ROC curve, with an exceptionally high true positive rate and minimal false positives, significantly outperforming all benchmark models.

### 5.2. Baseline comparison

To comprehensively assess the performance of the SCA-mechanism model, we employed standard evaluation metrics widely used in medical image classification. These metrics ensure a robust analysis of the model's effectiveness in distinguishing between lung and colon cancer cases.

Table 3 presents a comparative analysis of the SCA-mechanism model against six CNN benchmark architectures, evaluating Macro Precision, Macro Recall, Macro F1-Score and Accuracy across training and test sets.

The integration of separable convolutions and attention mechanisms significantly enhances feature extraction, leading to improved model generalization and diagnostic accuracy. The exceptional performance of SCA-mechanism highlights its potential as a highly reliable AI-driven tool for computationally efficient clinical applications in cancer detection.

**Table 1**
Layer-wise architecture of the proposed SCA-mechanism model and experimental details.

| Layer (Type) | Output shape | Parameters |
|---|---|---|
| InputLayer | (None, 224, 224, 3) | 0 |
| CustomSeparableConv2D | (None, 224, 224, 32) | 0 |
| BatchNormalization | (None, 224, 224, 32) | 128 |
| ReLU | (None, 224, 224, 32) | 0 |
| MaxPooling2D | (None, 112, 112, 32) | 0 |
| CustomSeparableConv2D | (None, 112, 112, 64) | 0 |
| BatchNormalization | (None, 112, 112, 64) | 256 |
| ReLU | (None, 112, 112, 64) | 0 |
| MaxPooling2D | (None, 56, 56, 64) | 0 |
| CustomSeparableConv2D | (None, 56, 56, 128) | 0 |
| BatchNormalization | (None, 56, 56, 128) | 512 |
| ReLU | (None, 56, 56, 128) | 0 |
| MaxPooling2D | (None, 28, 28, 128) | 0 |
| GlobalAveragePooling2D | (None, 128) | 0 |
| Dense | (None, 128) | 16,512 |
| Dense | (None, 128) | 16,512 |
| Reshape | (None, 1, 1, 128) | 0 |
| Multiply | (None, 28, 28, 128) | 0 |
| Flatten | (None, 100352) | 0 |
| Dense | (None, 256) | 25,690,368 |
| Dropout | (None, 256) | 0 |
| Dense | (None, 5) | 1,285 |
| **Training Details** | | |
| Augmentation | Horizontal flip (0.5), rotation (0.2), zoom (0.2) | – |
| Optimizer | Adam | – |
| Optimizer Hyperparameters | $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay $= 10^{-4}$ | – |
| Test time Augmentation | Yes (random crop and flip) | – |
| Learning Rate Schedule | Linear decay: $\eta_t = 0.001\ (1 - \frac{t}{T})$ | – |
| Initial Learning Rate | 0.001 | – |
| Batch Size | 32 | – |
| Epochs | 50 | – |

**Table 2**
Per-class accuracy and macro-average performance of the proposed model (SCA-mechanism).

| Class | Description | Accuracy (%) |
|---|---|---|
| colon_aca | Colon adenocarcinoma | 99.90 |
| colon_n | Normal colon tissue | 99.60 |
| lung_aca | Lung adenocarcinoma | 96.60 |
| lung_n | Normal lung tissue | 100.00 |
| lung_scc | Lung squamous cell carcinoma | 98.00 |
| **Macro Average (Balanced Accuracy)** | | **98.82** |

**Table 3**
Baseline of SCA-mechanism and benchmark models (mean ± sd).

| Model | Macro P (%) | Macro R (%) | Macro F1 (%) | Train Acc. (%) | Val Acc. (%) | Test Acc. (%) |
|---|---|---|---|---|---|---|
| ResNet50 | $97.88 \pm 0.12$ | $97.82 \pm 0.15$ | $97.82 \pm 0.14$ | $97.88 \pm 0.10$ | $97.60 \pm 0.18$ | $97.82 \pm 0.12$ |
| InceptionV3 | $66.44 \pm 0.55$ | $66.20 \pm 0.60$ | $60.25 \pm 0.70$ | $63.82 \pm 0.65$ | $62.80 \pm 0.70$ | $63.20 \pm 0.62$ |
| MobileNetV2 | $89.31 \pm 0.28$ | $89.31 \pm 0.26$ | $89.31 \pm 0.27$ | $86.55 \pm 0.35$ | $85.00 \pm 0.30$ | $85.38 \pm 0.32$ |
| DenseNet121 | $92.37 \pm 0.22$ | $91.74 \pm 0.24$ | $91.74 \pm 0.25$ | $89.90 \pm 0.30$ | $89.20 \pm 0.28$ | $89.76 \pm 0.26$ |
| EfficientNetB7 | $96.87 \pm 0.16$ | $96.76 \pm 0.17$ | $96.76 \pm 0.18$ | $96.10 \pm 0.20$ | $95.40 \pm 0.22$ | $95.74 \pm 0.20$ |
| VGG16 | $98.87 \pm 0.14$ | $98.67 \pm 0.15$ | $98.67 \pm 0.14$ | $98.18 \pm 0.16$ | $97.50 \pm 0.18$ | $97.67 \pm 0.17$ |
| **SCA-mechanism** | $\mathbf{98.82 \pm 0.12}$ | $\mathbf{98.80 \pm 0.12}$ | $\mathbf{98.82 \pm 0.11}$ | $\mathbf{99.33 \pm 0.07}$ | $\mathbf{98.60 \pm 0.13}$ | $\mathbf{98.82 \pm 0.10}$ |

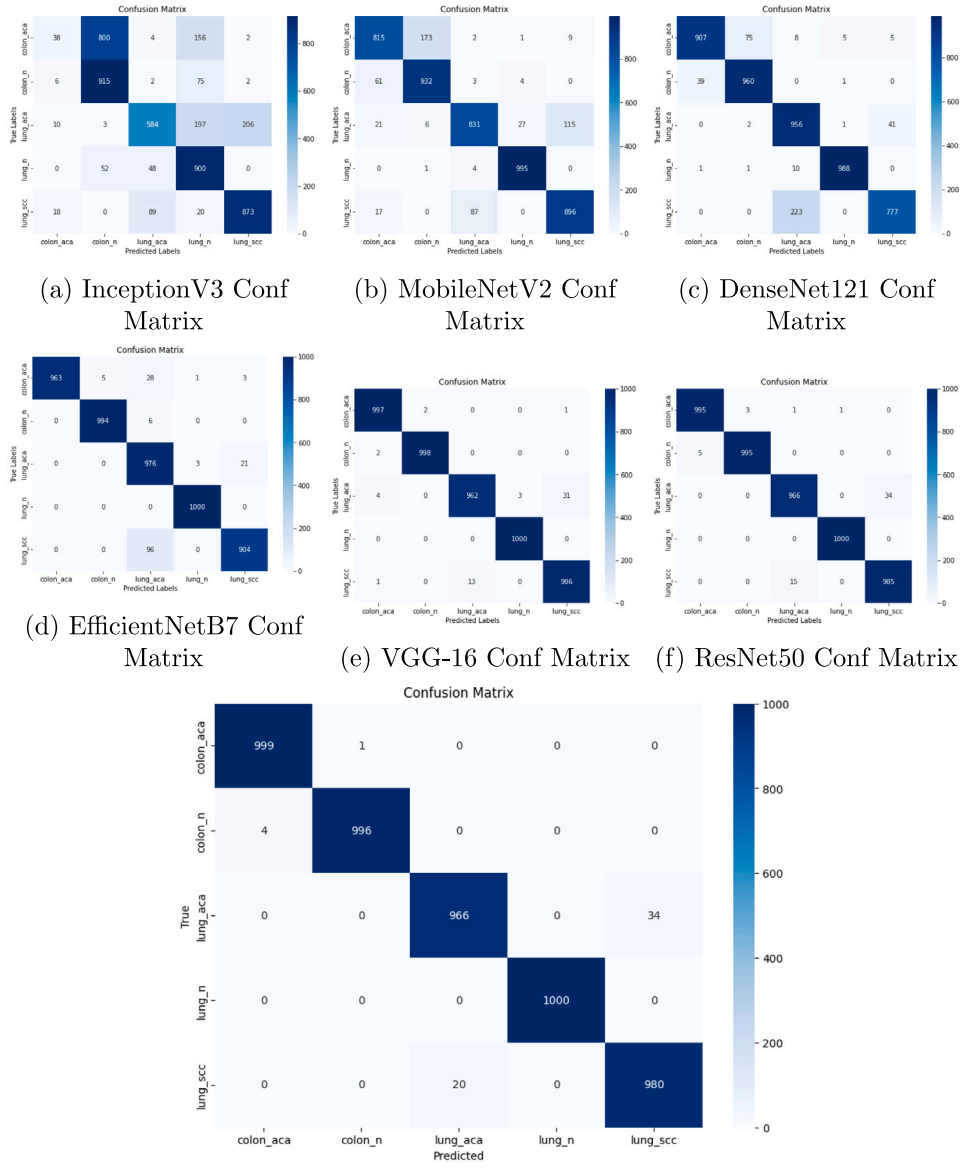
### *5.3. Explainable AI*

To ensure transparency and trust in deep learning models applied to histopathological image classification, we employed two widely adopted explainability techniques: Local Interpretable Model-Agnostic Explanations (LIME) and Gradient-weighted Class Activation Mapping (Grad-CAM). These methods were applied to histopathological images of colon and lung tissues–specifically for cancerous (Colon_aca) and non-cancerous (Colon_n, Lung_n) categories–to visualize and interpret the model's decision-making process. Fig. 5 illustrates the original images alongside the corresponding LIME and Grad-CAM explanations. Each row represents a different class of histopathological sample.

The LIME explanations highlight superpixels in yellow that most significantly influenced the model's prediction. LIME offers a local interpretability approach by perturbing input features and observing changes in the output, enabling an understanding of which regions the model focused on for a particular classification. The Grad-CAM heatmaps visualize class-discriminative regions in the convolutional layers by leveraging the gradients of the target class with respect to the feature maps. Regions with higher activation (warmer colors) indicate areas the model considers most relevant for the prediction.

From the visual comparisons:

- For cancerous tissues (e.g., Colon_aca), both LIME and Grad-CAM consistently highlight glandular and structural irregularities that are known to be pathological indicators.
- For normal tissues Colon_n and Lung_n, the highlighted areas are more uniformly distributed, aligning with expected tissue regularity.




(a) InceptionV3 Conf Matrix (b) MobileNetV2 Conf Matrix (c) DenseNet121 Conf Matrix

(d) EfficientNetB7 Conf Matrix (e) VGG-16 Conf Matrix (f) ResNet50 Conf Matrix

(g) SCA-mechanism Conf Matrix

**Fig. 3.** Confusion matrices of different models.

### 5.4. Average training time comparison

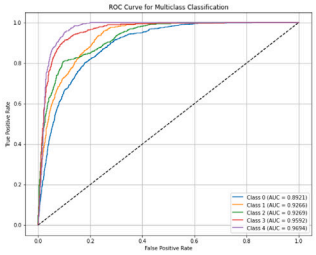
Fig. 6 provides a visual representation of the average training time comparison across different models. The computational efficiency of the SCA-mechanism model is significantly enhanced by the integration of Separable Convolutions with Residual Connections, reducing overall training time compared to state-of-the-art deep learning models. The proposed model completes each epoch in 102 s in average, demonstrating a substantial improvement in training speed.

The notable reduction in training time underscores the efficiency of the SCA-mechanism model, enabling faster convergence while maintaining high classification accuracy. This efficiency makes it particularly suitable for multiclass classification tasks in resource-constrained environments.

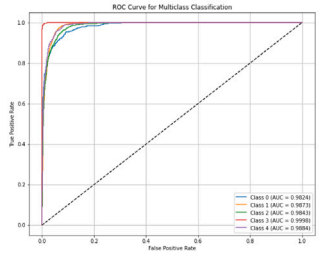
### 5.5. Ablation study and comparison

As shown in Table 4, the full SCA configuration (separable convolutions + residual connections + channel attention) achieves the best overall trade-off, reaching 98.82% ± 0.12 accuracy and 0.988 ± 0.002 macro-F1 with a compact footprint (25.69M params; 0.58G FLOPs) and low latency (2.5 ± 0.1 ms/img, batch = 1, 224 × 224). Relative to the Base CNN (96.1% ± 0.3; 0.960 ± 0.004), SCA improves accuracy by +2.8 pp and macro-F1 by +0.028; compared to SepConv-only (97.0% ± 0.2; 0.969 ± 0.003), it adds +1.9 pp and +0.019 for a modest compute increase (+0.10G FLOPs, ~ +0.4 ms/img). Component isolation indicates complementary benefits: adding residuals to separable blocks (SepConv+Residual, 97.6% ± 0.2) stabilizes optimization and improves depth, attention alone (SepConv+Attn, 97.2% ± 0.3) enhances channel selectivity, and combining both yields the most accurate and stable model.

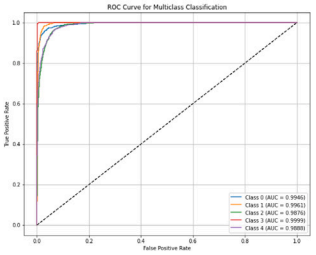
Design alternatives in Table 4 show expected trade-offs. A dilated mid-block (SCA–Dilated, rate = 2) slightly boosts contextual capture (98.5% ± 0.2) but increases compute/latency; swapping GAP for GMP (SCA+GMP, 98.3% ± 0.3, macro-F1 0.982 ± 0.003) reduces calibration/robustness, consistent with GMP's sensitivity to localized noise. Mild color jitter and focal loss yield small but consistent gains under stain and imbalance variability (≈98.6–98.7% accuracy), AdamW edges Adam



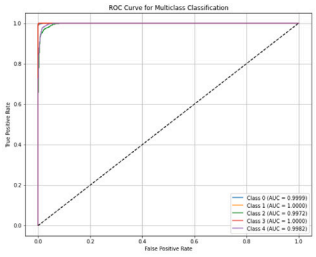

(a) InceptionV3 ROC Curve



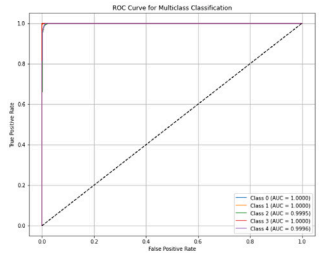

(b) MobileNetV2 ROC Curve



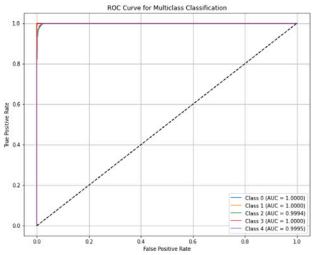

(c) DenseNet121 ROC Curve



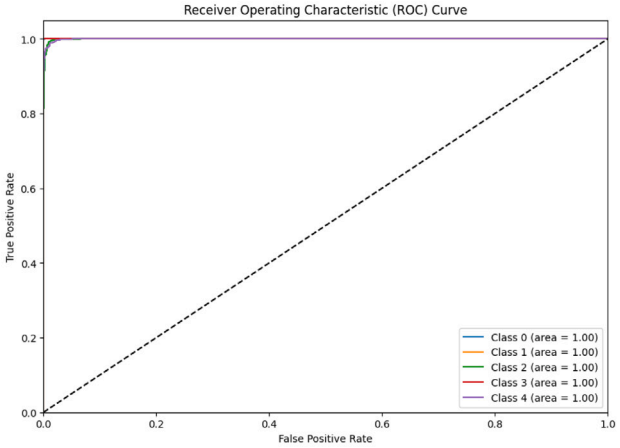

(d) EfficientNetB7 ROC Curve




(e) VGG-16 ROC Curve




(f) ResNet50 ROC Curve




(g) SCA-mechanism ROC Curve

**Fig. 4.** ROC curves of different models.

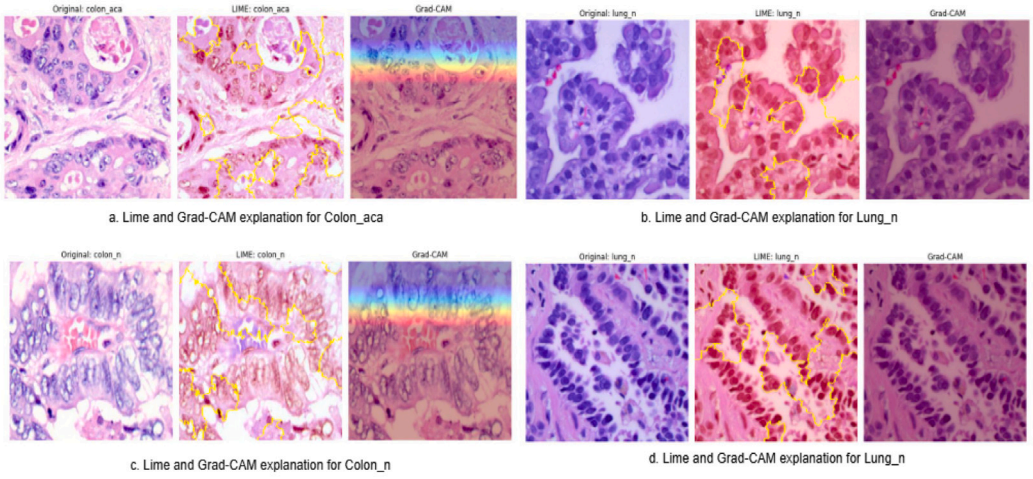



**Fig. 5.** Visual interpretability of deep learning predictions on histopathological images using LIME and Grad-CAM for Lung and Colon Cancer.

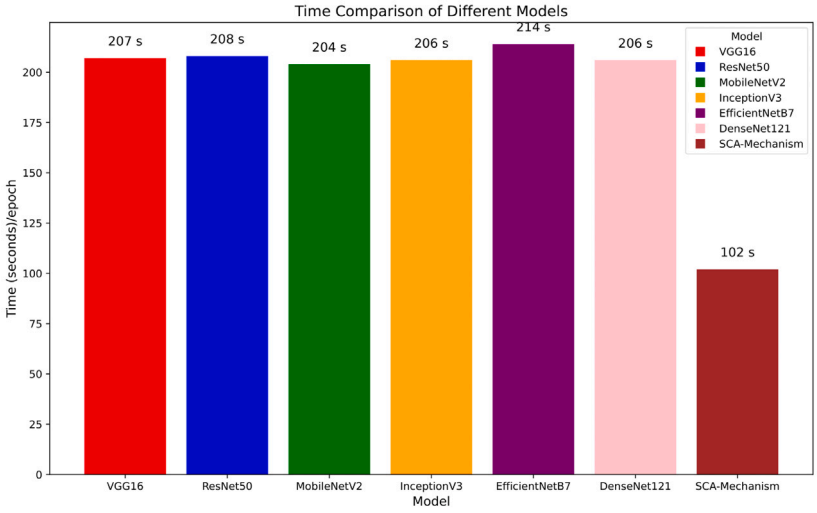



**Fig. 6.** Comparison Between training Time per step for the proposed Model with other different Models.

**Table 4**
Ablation study of SCA components and training choices.

| Model variant | Acc (%) | Macro F1 | Params (M) | FLOPs (G) | GPU (ms/img) |
|---|---|---|---|---|---|
| Base CNN (std conv, no res/attn) | $96.1 \pm 0.3$ | $0.960 \pm 0.004$ | 27.40 | 1.10 | $3.8 \pm 0.2$ |
| SepConv (no res/attn) | $97.0 \pm 0.2$ | $0.969 \pm 0.003$ | 25.10 | 0.48 | $2.1 \pm 0.1$ |
| SepConv + Residual (no attn) | $97.6 \pm 0.2$ | $0.975 \pm 0.003$ | 25.20 | 0.50 | $2.2 \pm 0.1$ |
| SepConv + Attn (no res) | $97.2 \pm 0.3$ | $0.972 \pm 0.004$ | 25.25 | 0.55 | $2.4 \pm 0.1$ |
| SepConv + Attn (Spatial) | $97.6 \pm 0.3$ | $0.98 \pm 0.2$ | 25.42 | 0.56 | $2.4 \pm 0.2$ |
| SepConv + Attn (Axial) | $98.76 \pm 0.4$ | $0.988 \pm 0.32$ | 25.50 | 0.55 | $2.45 \pm 0.2$ |
| **SCA (Channel Attn)** | $98.82 \pm 0.12$ | $0.988 \pm 0.002$ | 25.69 | 0.58 | $2.5 \pm 0.1$ |
| SCA–Dilated (rate = 2) | $98.5 \pm 0.12$ | $0.985 \pm 0.002$ | 25.38 | 0.64 | $2.8 \pm 0.1$ |
| SCA + GMP (vs GAP) | $98.3 \pm 0.3$ | $0.982 \pm 0.003$ | 25.30 | 0.58 | $2.5 \pm 0.1$ |
| SCA + Color Jitter (mild) | $98.6 \pm 0.2$ | $0.986 \pm 0.002$ | 25.30 | 0.58 | $2.5 \pm 0.1$ |
| SCA + Focal Loss ($\gamma = 2$) | $98.7 \pm 0.2$ | $0.987 \pm 0.002$ | 25.30 | 0.58 | $2.6 \pm 0.1$ |
| SCA + AdamW | $98.8 \pm 0.2$ | $0.987 \pm 0.002$ | 25.30 | 0.58 | $2.5 \pm 0.1$ |
| SCA + RMSprop | $98.2 \pm 0.3$ | $0.981 \pm 0.003$ | 25.30 | 0.58 | $2.6 \pm 0.1$ |
| SCA (dropout $p = 0.4$) | $98.7 \pm 0.2$ | $0.987 \pm 0.002$ | 25.30 | 0.58 | $2.5 \pm 0.1$ |

*Reporting:* mean ± SD over 3 seeds; selection on validation; test once. Batch= 1, 224 × 224. FLOPs computed as MACs × 2.

**Table 5**
Comparison of the SCA-mechanism model with existing methods for detecting lung and colon cancer in the LC25000 dataset.

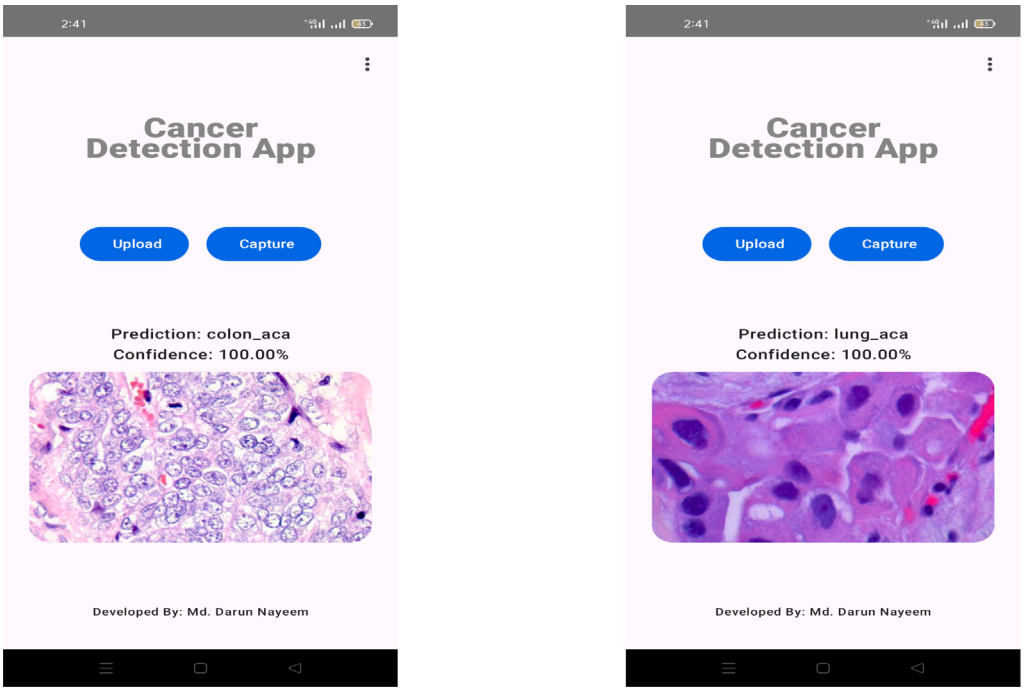
| Author | Cancer type | Image type | Model | Test accuracy (%) | Accuracy Diff. |
|---|---|---|---|---|---|
| Ibrahim et al. [35] | Lung | X-ray + CT | VGG19 + CNN | 98.05 | 0.77% ↓ |
| Provath et al. [36] | Lung | Histopathological | CNN | 97.00 | 1.82% ↓ |
| Bukhari et al. [37] | Colon | Histopathological | ResNet-50 | 93.91 | 4.91% ↓ |
| Da Nobrega et al. [38] | Lung | CT Scan | ResNet50 + SVM RBF | 93.19 | 5.63% ↓ |
| Carvalho Filho et al. [39] | Colon | Colonoscopy | AlexNet | 91.47 | 7.35% ↓ |
| Masud et al. [40] | Lung and Colon | Histopathological | CNN | 96.33 | 2.49% ↓ |
| **Proposed Model** | **Lung and Colon** | **Histopathological** | **SCA-mechanism** | **98.82** | – |

while RMSprop underperforms, and lowering dropout to $p = 0.4$ performs comparably (98.7%±0.2), indicating regularization robustness. Overall, SCA sits on a favorable Pareto front—near-minimal latency and model size with state-of-the-art accuracy that justify its use as the default configuration throughout.

Table 5 presents a comparative analysis of various deep learning and machine learning models employed in cancer diagnosis across multiple studies. It evaluates model performance based on different cancer types and imaging modalities, using key metrics such as Test Accuracy, Precision, Recall, and F1-Score. This comparison highlights the effectiveness of the SCA-mechanism model, which leverages Self-Attention Convolutional Mechanisms for lung and colon cancer detection in histopathological images. Notably, our model achieves a Test Accuracy of 98.82%, demonstrating a substantial improvement over previously published models. The results emphasize the impact of integrating separable convolutions with attention mechanisms, leading to enhanced feature extraction and superior classification performance.

### 5.6. Development of a mobile application

To support practical deployment and broad accessibility, the trained SCA mechanism was converted into a TensorFlow Lite (TFLite) model for lightweight on device inference. The mobile application was developed in Android Studio using Kotlin and XML to ensure a seamless and responsive user interface. The application allows users to capture new histopathological images or upload existing ones from their device. Once an image is provided, the model performs local inference without




(a) Prediction: colon_aca class (b) Prediction: lung_aca class

**Fig. 7.** User interface of the developed mobile cancer detection application.

requiring an internet connection, thereby preserving data privacy and ensuring low latency predictions. This offline and self-contained design makes the system particularly suitable for use in resource-limited or remote healthcare settings where stable network connectivity may be unavailable. An illustration of the application interface is shown in Fig. 7. Validation on unseen histopathological images of lung and colon cancer tissues demonstrated that the mobile application consistently produced accurate and high-confidence predictions. Overall, integrating the SCA model into a mobile environment bridges the gap between computational pathology research and practical diagnostic support.

## 6. Discussion and limitations

The results of this study highlight the strong potential of the SCA-mechanism as an effective and computationally efficient approach for lung and colon cancer classification. Across all evaluation metrics, the model demonstrates high accuracy, stable generalization, and reliable discrimination of malignant and benign tissue patterns. The combination of separable convolutions, residual connections, and channel attention contributes to consistent performance gains, confirming the value of integrating lightweight feature extraction with adaptive feature prioritization. The ablation study further reinforces the robustness of the architecture, showing that the full SCA configuration offers a favorable balance between predictive power, model size, and inference speed. The explainability analysis with LIME and Grad-CAM also shows that the model consistently focuses on clinically meaningful regions, strengthening confidence in its decision-making process. Moreover, the successful conversion to a TensorFlow Lite format and deployment within a mobile application illustrate the practical feasibility of using the model in real world scenarios, particularly where local inference and data privacy are essential.

Although the findings are highly encouraging, certain considerations remain for future development. The model has been trained and evaluated on the LC25000 dataset, which provides high quality and balanced samples but does not encompass the full range of variability encountered in diverse clinical settings. Additional validation on multi institutional datasets and different imaging conditions would help reinforce the model's general applicability. Similarly, the use of $224 \times 224$ resized images, while beneficial for computational efficiency, may limit the capture of extremely fine morphological details. Incorporating multi scale feature extraction techniques could further enhance performance. Finally, while the current work focuses on image level classification, extending the framework to region level or slide level analysis may broaden its clinical utility.

Overall, the SCA mechanism delivers strong and consistent performance, efficient computation, and promising interpretability, positioning it as a viable candidate for real world computational pathology applications and future clinical integration.

## 7. Conclusions and future works

This study introduced the SCA-mechanism model, a novel hybrid architecture integrating separable convolutions and attention mechanisms for lung and colon cancer classification. By enhancing feature extraction and prioritizing critical image regions, the model significantly improved classification accuracy, and computational efficiency. The SCA-mechanism model outperformed traditional deep learning models, surpassing DenseNet121, EfficientNetB7, ResNet50, and others, achieving a 98.82% accuracy. This hybrid architecture represents a step forward in AI-driven cancer detection, with the potential to enhance clinical decision-making and patient outcomes.

While the SCA-mechanism model demonstrates strong performance, there are areas for further improvement. Currently, it has been trained on the LC25000 dataset, and additional testing on diverse datasets, including other imaging modalities like radiographic or endoscopic images, would enhance its generalizability.

## CRediT authorship contribution statement

**Md. Darun Nayeem:** Writing – original draft, Methodology, Formal analysis, Conceptualization. **Md. Emdadul Hasan Shishir:** Writing – original draft, Formal analysis, Data curation. **Munshi Touibur Rahman:** Writing – original draft, Investigation, Formal analysis, Data curation. **Zeeshan Chowdhury Juwel:** Visualization, Formal analysis,

Data curation. **Sagor Sutradhar:** Visualization, Formal analysis, Data curation. **Sudipto Chaki:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Formal analysis. **Md. Saifur Rahman:** Writing – review & editing, Resources, Project administration. **A.B.M. Shawkat Ali:** Resources, Project administration, Funding acquisition.

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## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Data availability

Lung and Colon Cancer Histopathological Image Dataset (LC25000) Available at https://shorturl.at/xJmR5.

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